

考研精读 | The Guardian



Pet detectives: dogs can tell fearful from angry or sad human faces, research finds

萌宠大侦探：研究发现，狗狗可以识别人脸愤怒或悲伤的情绪

难度评级：★★☆☆☆

【 Para.1 】

While humans **keep a close eye on** their pets, it seems our pets are keeping a close eye on us too, as a study has revealed dog brains can **distinguish** between different types of unhappy human expressions.

虽然人类会密切关注自己的宠物，但似乎我们的宠物也在密切关注着我们——一项研究表明，狗的大脑能够区分人类不同类型的不开心表情。

【 Para.2 】

Dogs whose brain activity was **monitored** not only distinguished happy faces from other expressions, but could tell fearful faces apart from those that are angry or sad, researchers found.

研究人员通过监测狗狗的大脑活动，发现它们不仅能

11 August 2026 | The Guardian

文章导读：

维也纳大学借助核磁共振开展实验研究，发现狗狗能够识别人类开心的面部表情。依托全脑神经活动，犬只可以区分恐惧表情与悲伤、愤怒表情，但难以分辨愤怒和悲伤。该研究同样存在不足之处，主要是样本存在局限。本研究证实驯化塑造了犬理解人类情绪的能力，相关结论对训犬具备实用价值。

detective n. A detective is someone whose job is to discover what has happened in a crime or other situation and to find the people involved. Some detectives work in the police force and others work privately. 侦探【考研英语大纲词汇】

keep an eye on sb/sth phrase to take care of sb/sth and make sure that they are not harmed, damaged, etc. 照看；留神；留意

We've asked the neighbours to **keep an eye on the house** for us while we are away. 我们已请邻居在我们离开时帮我们照看一下房子。

distinguish v. ~ (between) A and B | ~ A from B to recognize the difference between two people or things 区分；辨别；分清【考研英语大纲词汇】

SYN differentiate

At what age are children able to **distinguish** between right and wrong? 儿童到什么年龄才能明辨是非？

monitor v. to watch and check sth over a period of time in order to see how it develops, so that you can make any necessary changes 监视；检查；跟踪调查【考研英语大纲词汇】

SYN track

Each student's progress is closely **monitored**. 每一位同学的学习情况都受到密切的关注。

demonstration n. an act of giving proof or evidence for sth 证明；证实；论证；说明【考研英语大纲词汇】

valence n. 效价——情感体验的积极/消极程度(正面/负面)

unrestrained a. /ˌʌnrɪ'streɪnd/ (formal) not controlled; not having been restrained 无节制的；放纵的；不加制约的

caregiver n. (=carer) a person who takes care of a sick or old person at home 照料家居老弱病患者的人；家庭护理员

bear v. ~ sth (against/towards sb) | ~ sb sth to have a feeling, especially a negative feeling 心怀(感情，尤指坏心情)【考研英语大纲词汇】

He **bears** no resentment towards them. 他对他们毫无怨恨。

trigger v. ~ sth (off) to make sth happen suddenly 发动；引起；触发【考研英语大纲词汇】

SYN set off

Nuts can **trigger** off a violent allergic reaction. 坚果可以引起严重的过敏反应。

区分快乐或其他面部表情，还能区分恐惧、愤怒或悲伤的面部表情。

tend v. to be likely to do sth or to happen in a particular way because this is what often or usually happens 往往会；常常就【考研英语词汇】

Women **tend** to live longer than men. 女人往往比男人长寿。

collie n. a dog of which there are several types. Those with long pointed noses and long thick hair are popular as pets. Smaller collies with shorter hair are often trained to help control sheep on a farm. 科利牧羊狗

implication n. /ˌɪmplɪˈkeɪʃn/ [C, usually pl.]~ (for/of sth) a possible effect or result of an action or a decision 可能的影响（或作用、结果）【考研英语词汇】

The development of the site will have **implications** for the surrounding countryside. 这个地点的开发将会影响周围的乡村。

domestication n. (domestic+ -ation) adaptation to intimate association with human beings 驯化

make sense phrase to have a meaning that you can easily understand 有道理；有意义；讲得通【考研英语词汇】

This sentence doesn't **make sense**. 这个句子不通。

【 Para.3 】

Dr Laura Cuaya said: “This is the first **demonstration** that the dog brain can distinguish between emotions from the same **valence** [positive or negative character],” she said.

劳拉·库亚博士表示：“这项研究首次证明狗的大脑能够区分同一情感效价（积极或消极）下的不同情绪。”

【 Para.4 】

Writing in the journal iScience, Cuaya and colleagues from the University of Vienna described working with pet dogs that had been trained to lie still, awake and **unrestrained**, within an MRI scanner.

库亚及其维也纳大学的同事将他们的研究发现刊登在《iScience》期刊上。文章描述了他们与经过训练的宠物狗进行实验的过程——这些狗可以在磁共振成像（MRI）扫描仪内保持静止、清醒且不受束缚的状态。

【 Para.5 】

In the first experiment, eight dogs were shown five sets of happy faces and five sets of neutral faces. Each set contained four photos of humans that were unfamiliar to the dog but were of the same gender as their primary **caregiver**. This was repeated five times for each dog.

在第一项实验中，研究人员向8只狗展示了5组快乐的面孔和5组中性情绪的面孔。每组包含4张人类照片，这些照片对狗来说是陌生的，但与它们的主要照料者属于同一性别。每只狗都重复进行了5次该实验。

【 Para.6 】

The results revealed increased activity in certain parts of the brain when the dogs were shown happy faces compared with neutral ones. Cuaya said these areas were associated with higher visual processing, social processing and reward processing – suggesting the dog was experiencing something positive.

结果显示，与中性表情相比，当向狗展示快乐表情时，其大脑某些区域的活动增强。库亚表示，这些区域与高级视觉处理、社交处理和奖励处理相关——这表明狗正在体验某种积极的情绪。

【 Para.7 】

Dr Raúl Hernández Pérez, the first author of the study, said the results raised another question, however: “We don’t know if these regions are related just to happy faces or to emotional faces in general.”

不过，该研究的第一作者劳尔·埃尔南德斯·佩雷斯博士表示，这一结果也引发了另一个问题：“我们尚不清楚这些脑区是仅与快乐的面孔相关，还是与所有带情绪的面孔相关。”

今日句子分析

Para 2. Dogs whose brain activity was monitored not only distinguished happy faces from other expressions, but could tell fearful faces apart from those that are angry or sad, researchers found. 研究人员通过监测狗狗的大脑活动，发现它们不仅能区分快乐或其他面部表情，还能区分恐惧、愤怒或悲伤的面部表情。

宾语从句	定语从句	
Dogs	whose brain activity	was monitored
主语	从句主语	谓语
not only distinguished	happy faces	from other expressions,
谓语	宾语	状语
but could tell	fearful faces	apart from those
(连词)	谓语	宾语 状语
定语从句	主句	
that	are angry or sad,	researchers found.
主句	系语	表语 主句 谓语

句子分析：

主句：

主句-researchers；谓语-found（这种用于说明出处的主谓，sb says/reckons/reports/...类同，都可以较为灵活地在句中游走）；

宾语从句：

主句-Dogs；谓语-not only distinguished（将.....与.....区分开来）；宾语-happy faces；状语-from other expressions；

（连词）-but；谓语-could tell；宾语-fearful faces；状语-apart from those（tell apart同理，加个from则明确区分对象）；

定语从句，修饰dogs：

从句主语-whose brain activity；谓语-was monitored；

定语从句，修饰fearful faces：

从句主语-that；系语-are；表语-angry or sad。

翻译建议：Dogs whose brain activity was monitored根据原句翻译，或许会范围“研究人员发现，那些大脑活动被监测的狗不仅能.....”，但实际上这么翻容易引起歧义，好似“只有在大脑活动被监测的状态下，狗狗才能.....”。为了避免这种误读，可以进一步思考下whose brain activity was monitored的作用，它其实是为了说明研究人员用什么手段得出这一结论，所以在翻译时，可以考虑把这一部分从原先的定语转化为方式状语，进行翻译，即“研究人员通过监测狗狗的大脑活动.....”。

【Para.8】

The team then carried out a second MRI experiment involving 12 dogs. Each was shown new sets of unfamiliar male and female human faces, this time bearing four different expressions: happiness, sadness, fear and anger.

随后，研究团队对12只狗进行了第二次MRI实验。每只狗都被展示了新的、陌生的男性和女性人类面孔，这次的面孔呈现了四种不同的表情：快乐、悲伤、恐惧和愤怒。

【Para.9】

The researchers used machine learning to analyse activity in the same brain regions identified in the first experiment, revealing that the animals could distinguish happy faces from each negative expression – but could not distinguish between different types of unhappy face.

研究人员利用机器学习技术分析了与首次实验中相同的脑区活动，结果显示，参与实验的狗狗能够将快乐的面孔与每种负面表情区分开来——但无法区分不同类型的不快乐面孔。

【Para.10】

However, when activity across the whole dog brain was analysed, the researchers found it was possible to distinguish between sad v fearful faces, and angry v fearful faces – but not between angry and sad faces.

然而，当分析狗整个大脑的活动时，研究人员发

➤ 写作积累

Para.1 While humans keep a close eye on their pets, it seems our pets are keeping a close eye on us too, as a study has revealed dog brains can distinguish between different types of unhappy human expressions.

keep an eye on sb/sth 有“照看；监管”的意思，加上 close，则加了“仔细、严密（看管）”的意思，因此文中这里 humans keep a close eye on their pets 可以理解为“细心照料；密切观察”的意思，后半句则指“宠物其实也在密切关注主人”。

由此衍生的短语还包括：

keep an eye open/out for sb/sth “密切留意；提防”，这里的 for 并非指“为了”，而是指“寻找”，例：Police have asked residents to keep an eye out for anything suspicious. 警方要求居民密切注意一切可疑的情况。

而与 keep an eye on 几乎可以互换的短语有 watch over sb/sth “照看；关注……”，例：

These mothers take turns to watch over each other's children. 这些母亲轮流照看彼此的孩子。

Para.13 The authors said it might be harder for dogs to tell apart two human facial expressions if they were both negative or both positive, and dogs might also need additional information – such as body language – to distinguish anger from sadness.

文中有多处出现“区分；区别”的动词词组，使用起来略有不同：

1. tell apart “区分”一般是两个不同种类中进行区分，较为日常口语，例：The two sides can be hard to tell apart. 双方可能很难分辨。

2. distinguish (differentiate 同理) A from B, “将 A 与 B 区分开来”（侧重点会落在 A），这个搭配下也是两种区分，例：Sometimes reality and fantasy are hard to distinguish. 有时候现实和幻想很难区分。

distinguish between A and B 则偏向于平等地“区别 A 与 B”，例：At what age are children able to distinguish between right and wrong? 儿童到什么年龄才能明辨是非？

现，虽然它们可以区分悲伤的面孔与恐惧的面孔，以及愤怒的面孔与恐惧的面孔——但无法区分愤怒的面孔与悲伤的面孔。

【Para.11】

Hernández said different negative expressions may trigger different responses in dogs as they represented different types of threat.

埃尔南德斯表示，不同的负面表情可能在狗身上引发不同的反应，因为它们代表了不同类型的威胁。

【Para.12】

Cuaya said previous behavioural studies had shown dogs can tell apart some human facial expressions, but that they tended to only test two at a time, one positive and one negative.

库亚指出，此前行为学研究已表明狗能够区分某些人类面部表情，但这些研究通常每次只测试两种表情，即一种积极表情和一种消极表情。

【Para.13】

The authors said it might be harder for dogs to tell apart two human facial expressions if they were both negative or both positive, and dogs might also need additional information – such as body language – to distinguish anger from sadness.

研究作者指出，如果两种人类面部表情均为负面或均为正面，狗可能更难加以区分；此外，狗可能还需要额外的信息——例如肢体语言——才能将愤怒与悲伤区分开来。

【Para.14】

The research has limitations, including its small sample size; the fact that most of the dogs were border collies, and all were healthy and without behavioural problems.

该研究存在一些局限性，包括样本量较小；参与实验的狗大多是边境牧羊犬，且均为健康个体，没有行为问题。

【Para.15】

Cuaya said the study could have practical implications: for example, expressing happiness when training a dog could aid the learning process.

库亚表示，这项研究可能具有实际意义：例如，在训练狗时表现出快乐的情绪，可能会有助于整个训练过程。

【Para.16】

Bridget Waller, a professor of evolution and social behaviour at Nottingham Trent University, who was not involved in the study, described the work as fascinating, saying it showed how “dogs have been

shaped through domestication to understand human behaviour”.

诺丁汉特伦特大学进化与社会行为学教授布里奇特·沃勒 (Bridget Waller) 虽未参与该研究, 但她称这项研究非常吸引人, 并表示它表明 “狗在驯化过程中逐渐发展出了理解人类行为的能力”。

【Para.17】

“Smiling is the most commonly used human facial expression, vital to our social interactions, and so it

makes sense that dogs have developed a particular ability to distinguish this from other facial movements,” she said. “It seems less clear that they can distinguish between the other expressions, and this would make sense as these are less commonly produced in human social interaction.”

“微笑是人类最常用的面部表情, 对我们的社交互动至关重要, 因此狗狗发展出一种特殊能力来将微笑与其他面部动作区分开来, 这是合乎情理的, ” 她说。 “至于它们能否区分其他表情, 目前尚不明确, 但这也合理, 因为这些表情在与人类社交互动中出现的频率较低。”